

Linear Algebra HW 7

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Question 1

(a)

$$\begin{aligned}\chi_A(\lambda) &= \det(A - \lambda I) \\ \chi_A(\lambda) = 0 &\iff (-5 - \lambda)(-5 - -\lambda) = 0\end{aligned}$$

Thus we see that $\lambda = -5$ is the only eigenvalue of A

$\mathcal{N}(A - 5I)$:

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 5 & 0 & 0 \end{array} \right]$$

The dimension of this is 1, thus there exists only one linearly independent eigenvector of A , e.g. $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Thus we know that the Jordan form must be $\begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$

(b)

Now looking in the next nullspace $\mathcal{N}(A - 5I)^2$:

$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

One can clearly see that this space is all of $\mathbb{R}^2$, picking one of these such that is linearly independent from our earlier eigenvector, we get $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

This set $\mathcal{C} = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ then forms a Jordan basis for A

Thus $M = P_{\mathcal{C} \rightarrow \mathcal{E}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Question 2

(a)

$$\begin{aligned}\chi_A(\lambda) &= \det(A - \lambda I) \\ \chi_A(\lambda) = 0 &\iff (4 - \lambda)((-1 - \lambda)(-4 - \lambda) - 3) - 3(7(-4 - \lambda) + 24) + 4(7 + 8(-1 - \lambda)) \\ &\iff -\lambda^3 - \lambda^2 + 8\lambda + 12 = 0 \\ &\iff (\lambda - 3)(\lambda + 2)^2 = 0\end{aligned}$$

Thus we see that $\lambda = -2$ and $\lambda = 3$ are the two eigenvalues of A

$\mathcal{N}(A - 3I)$:

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 0 \\ 7 & -4 & 3 & 0 \\ -8 & 1 & -7 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The set of eigenvectors associated with $\lambda = 3$ is then $\left\{ \begin{bmatrix} -t \\ -t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\}$

$\mathcal{N}(A + 2I)$:

$$\left[\begin{array}{ccc|c} 6 & 3 & 4 & 0 \\ 7 & 1 & 3 & 0 \\ -8 & 1 & -2 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{3} & 0 \\ 0 & 1 & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The set of eigenvectors associated with $\lambda = -2$ is then $\left\{ \begin{bmatrix} -\frac{t}{3} \\ -\frac{2t}{3} \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\}$

$\mathcal{N}(A + 2I)^2$

$$\left[\begin{array}{ccc|c} 25 & 25 & 25 & 0 \\ 25 & 25 & 25 & 0 \\ -25 & -25 & -25 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The set of generalised eigenvectors associated with $\lambda = -2$ is then $\left\{ \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} \mid s, t \in \mathbb{R} \right\}$

Thus the generalised eigenspace associated with $\lambda = 3$ is $V(3) = \left\{ \begin{bmatrix} -t \\ -t \\ t \end{bmatrix} \mid t \in \mathbb{R} \right\}$

and the generalised eigenspace associated with $\lambda = -2$ is $V(-2) = \left\{ \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} \mid s, t \in \mathbb{R} \right\}$

Note: We do not need to include $\mathcal{N}(A + 2I)$ in the second eigenspace since it is contained within the set of generalised eigenvectors

(b)

Since A has two eigenvectors associated with $\lambda = 3$ and $\lambda = -2$, and one generalised eigenvector associated with $\lambda = -2$, we know that the Jordan form of A must be (up to permutations of the blocks)

$$J = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}$$

(c)

Picking one eigenvector associated with $\lambda = 3$ such as $\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$, then one generalised eigenvector associated

with $\lambda = -2$ such that it is linearly independent of the set of eigenvectors associated with $\lambda = -2$ such as $\begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, then our final vector in the Jordan basis is $(A + 2I)$ multiplied to the generalised eigenvector which

comes out to be $\begin{bmatrix} -5 \\ -10 \\ 15 \end{bmatrix}$

These three vectors then form a Jordan basis

Then we know that M has as its columns the images of these basis vector, hence

$$M = \begin{bmatrix} -1 & -5 & -2 \\ -1 & -10 & 1 \\ 1 & 15 & 1 \end{bmatrix}$$

Question 3

To determine the Jordan form of A , we must first look in the null spaces of $(A - 5I)^k$, $k \in \mathbb{N}$

$\mathcal{N}(A - 5I)$:

$$\begin{bmatrix} 0 & -3 & -2 & | & 0 \\ 1 & -5 & -3 & | & 0 \\ -1 & 8 & 5 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -5 & -3 & | & 0 \\ 0 & -3 & -2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

We see that $\dim(\mathcal{N}(A - 5I)) = 1$, since there is only one free variable, thus there is only linearly independent eigenvector of A , thus the Jordan form of A must take the form

$$J = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{bmatrix}$$

since any other Jordan form would require at least two eigenvectors

Question 4

We know every Jordan matrix is of the form $J = M^{-1}AM$, where M is an invertible matrix

Thus $A = MJM^{-1}$

Picking $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, we get

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -7 & 1 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} -7 & 1 & -1 \\ 0 & -7 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 5

Since the matrix A is a 3×3 matrix with three distinct eigenvalues we know that it's Jordan form will be

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Also A is of the form $A = MJM^{-1}$ where the columns of M are the eigenvectors of A with respect to the standard basis vectors,

$$\text{i.e. } M = \begin{bmatrix} 2 & 0 & 3 \\ 3 & -1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$$

$$\text{Then } M^{-1} = \begin{bmatrix} 2 & 0 & -3 \\ 7 & -1 & -11 \\ -1 & 0 & 2 \end{bmatrix}$$

Finally,

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 3 & -1 & -1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 & -3 \\ 7 & -1 & -11 \\ -1 & 0 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 8 & 0 & -12 \\ -23 & 5 & 37 \\ 4 & 0 & 6 \end{bmatrix}$$

Question 6

Let T and S be linear operators defined on the vector space V , such that $S \circ T = T \circ S$

Let $\vec{k} \in \ker(S)$ i.e. $S(\vec{k}) = \vec{0}$

Then since T is a linear operator we have that $T(S(\vec{k})) = \vec{0}$ and hence $S(T(\vec{k})) = \vec{0}$

i.e. $T(\vec{k}) \in \ker(S) \forall k \in \ker(S)$

Hence $\ker(S)$ is invariant under T

Question 7

Let $\mathcal{B} = \{1, x, x^2, x^3, x^4\}$ be a basis for $\mathbb{P}_4[x]$

Then $[T]_{\mathcal{B}, \mathcal{B}}$ is the matrix which has as its columns the images of these basis vectors

$$[T]_{\mathcal{B}, \mathcal{B}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Since $[T]_{\mathcal{B}, \mathcal{B}}$ is a diagonal matrix, we know that the eigenvalues of T are the entries along the diagonal, 0, 1, 2, 3, 4 and the eigenvectors are the spans of each of the vectors in $\mathcal{B}$